Algorithm 1: Solver dispatch. The dispatcher checks for a named solver first, then—only when the caller passed `auto`—for separability, and otherwise returns the trust-region family. An unrecognised solver name is an error, never a silent fallback, so a typo cannot quietly change what was measured.

Input : model graph G, solver request s
Output: a solver instance, or an error

```
if s names a solver then
    return that solver, or Error if the name is unknown
    // never a silent fallback
else
    if G is separable in its linear coefficients, and every varying nonlinear
      parameter is unbounded, and no parameter is tied, and no node is bound to
      a dataset index then
        return VariableProjection
        // the linear block is eliminated
    else
        return TrustRegionLevenbergMarquardt
```